

Miguel Villasis Miranda

Unreal Engine Developer

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Vancouver, BC, Canada

PROFESSIONAL SUMMARY

Unreal Engine Developer with over 5 years of experience architecting scalable pipeline tools, integrating open standards, and expanding engine capabilities in C++ and Python. Passionate about solving tricky interoperability problems, optimizing DCC-to-Engine workflows, and building robust, high-performance frameworks that preserve artistic intent and metadata across massive production pipelines.

PROFESSIONAL EXPERIENCE

Mainframe Studios, Vancouver, BC — Game Engine Pipeline TD

SEPTEMBER 2022 - PRESENT

High-Performance Asset Architecture: Architected and deployed a custom C++ "Shot Container" Data Asset system within Unreal Engine to manage data for thousands of production shots. Built Python bridging utilities to query external repositories, caching data within the engine to eliminate redundant Python execution, optimize memory budgets, and ensure real-time performance.

DCC -> Engine Interoperability & USD: Expanded and maintained custom Unreal Engine importers and exporters for USD and FBX. Handled complex 3D coordinate system conversions and matrix transformations between DCCs (Maya, Nuke, Gaffer) and Unreal, preserving scene structure and artistic intent.

Pipeline Optimization: Accelerated the import process for massive FBX backgrounds into the Engine by 200%, significantly reducing technical barriers for artists.

Engine Source Modification & Custom Builds: Compiled, managed, and distributed custom Unreal Engine source builds. Implemented C++ source code modifications and high-level debugging to resolve critical bottlenecks and tailor the engine to strict production constraints.

Cinematics Automation: Engineered tools using C++, Python, and Blueprints to enhance the Level Sequencer and Movie Render Queue, fully automating the import and rendering pipeline.

Environment & Automation Tooling: Configured robust development environments using Rez, Batch, and Bash on Windows, writing cross-discipline documentation and partnering directly with Supervisors and Tech Art to integrate tools with minimal production disruption.

Mainframe Studios, Vancouver, BC — UE Jr Pipeline TD

JANUARY 2021 - AUGUST 2022

Maintained and upgraded core Unreal Engine pipeline tools and plugins using C++, Python, and Blueprints, ensuring stability across evolving project needs.

Partnered with artists and designers to conceptualize, script, and deploy quality-of-life tools, establishing an early foundation in cross-discipline technical support.

SKILLS

Programming Languages:

C++ (17+), Python, C#.

Engines & DCCs:

Unreal Engine 5, Maya, Unity.

Pipeline & Standards:

USD, FBX, Custom Schemas, Rez.

Version Control:

Perforce, Git.

Frameworks:

UMG, Slate, Blueprints, Movie Render Queue.

PROJECTS

Unworthy Roses [↗](#)

Co-op action/adventure game made in Unreal Engine 4. Worked as project manager and gameplay designer/programmer.

Designed and implemented puzzle cooperative mechanics, combat systems, and progression features.

A Game Of Dragon [↗](#)

A 3D Third-Person, Casual Adventure game where you play as Barnon, a baby dragon in search of his older brothers.

Made in Unreal Engine 5.

Worked as a programmer.

LANGUAGES

English: Full Professional Proficiency (IELTS 8.5)

Spanish: Native Speaker

Poktapok Studios, Mexico City — *Game Development Intern*

SEPTEMBER 2018 - JANUARY 2019

Assisted in core gameplay programming and technical troubleshooting for games using Unity and C#.

EDUCATION

Vancouver Film School, Vancouver, BC — *Game Design Diploma (Honours)*

AUGUST 2019 - AUGUST 2020

Focus: Game Programming, Unreal Engine, Unity, Game Production

Universidad Panamericana, Mexico City — *Post-Degree in Graphics Simulation (Honours)*

AUGUST 2018 - JUNE 2019

Focus: Unreal Engine, Computer Graphics, XR, Haptics

Universidad Panamericana, Mexico City — *B.S. in Digital Animation Engineering*

AUGUST 2014 - JUNE 2018

Studies of Computer Science, Engineering, Game Development